

MySQL Notes: CRUD & ALTER TABLE

Complete notes for creating tables, inserting, reading, updating, deleting records, and modifying table columns.

1. CREATE TABLE

The **CREATE TABLE** statement is used to create a new table and define its columns and constraints.

```
CREATE TABLE Employees (  
  Emp_ID INT PRIMARY KEY,  
  Name VARCHAR(50),  
  Department VARCHAR(50),  
  Salary DECIMAL(10,2)  
);
```

2. INSERT DATA — Create Records

The **INSERT INTO** statement adds new records (rows) to a table.

```
INSERT INTO Employees (Emp_ID, Name, Department, Salary)  
VALUES  
(101, 'Rahul', 'IT', 45000),  
(102, 'Priya', 'HR', 40000),  
(103, 'Amit', 'Sales', 50000);
```

3. READ DATA

The **SELECT** statement is used to retrieve data from a table.

Read all columns and records

```
SELECT * FROM Employees;
```

Read specific columns

```
SELECT Name, Salary  
FROM Employees;
```

Read records using a condition

```
SELECT *  
FROM Employees  
WHERE Department = 'IT';
```

4. UPDATE DATA

The **UPDATE** statement modifies existing records. Always use an appropriate **WHERE** condition when you want to update specific rows.

```
UPDATE Employees  
SET Salary = 55000  
WHERE Emp_ID = 103;
```

5. DELETE DATA

The **DELETE** statement removes existing records. A missing WHERE clause can delete all rows.

```
DELETE FROM Employees  
WHERE Emp_ID = 102;
```

6. ALTER TABLE — Modify Table Structure

Use **ALTER TABLE** when you want to add, delete, or modify columns in an existing table.

Add a Column

```
ALTER TABLE Employees
ADD Email VARCHAR(100);
```

Add Multiple Columns

```
ALTER TABLE Employees
ADD Phone VARCHAR(15),
ADD City VARCHAR(50);
```

Modify a Column

Use **MODIFY** to change the data type, size, or other column definition.

```
ALTER TABLE Employees
MODIFY Salary DECIMAL(10,2);
```

Rename and Modify a Column

Use **CHANGE** when you want to rename a column and specify its definition.

```
ALTER TABLE Employees
CHANGE Name Full_Name VARCHAR(100);
```

Delete a Column

```
ALTER TABLE Employees
DROP COLUMN Email;
```

7. Quick Cheat Sheet

Task	SQL Command	Purpose
Create table	CREATE TABLE	Creates table structure
Insert records	INSERT INTO	Adds new rows
Read	SELECT	Retrieves data
Update	UPDATE	Modifies existing rows
Delete records	DELETE	Removes rows
Add column	ALTER TABLE ... ADD	Adds a column
Modify column	ALTER TABLE ... MODIFY	Changes column definition
Rename column	ALTER TABLE ... CHANGE	Renames and defines a column
Delete column	ALTER TABLE ... DROP COLUMN	Removes a column

Important Notes

- **CREATE TABLE** is DDL (Data Definition Language).
- **INSERT**, **UPDATE**, **DELETE** are DML (Data Manipulation Language).
- **SELECT** is used to query/read data.
- **ALTER TABLE** changes the structure of an existing table.
- Be careful with **UPDATE** and **DELETE** without a WHERE clause.